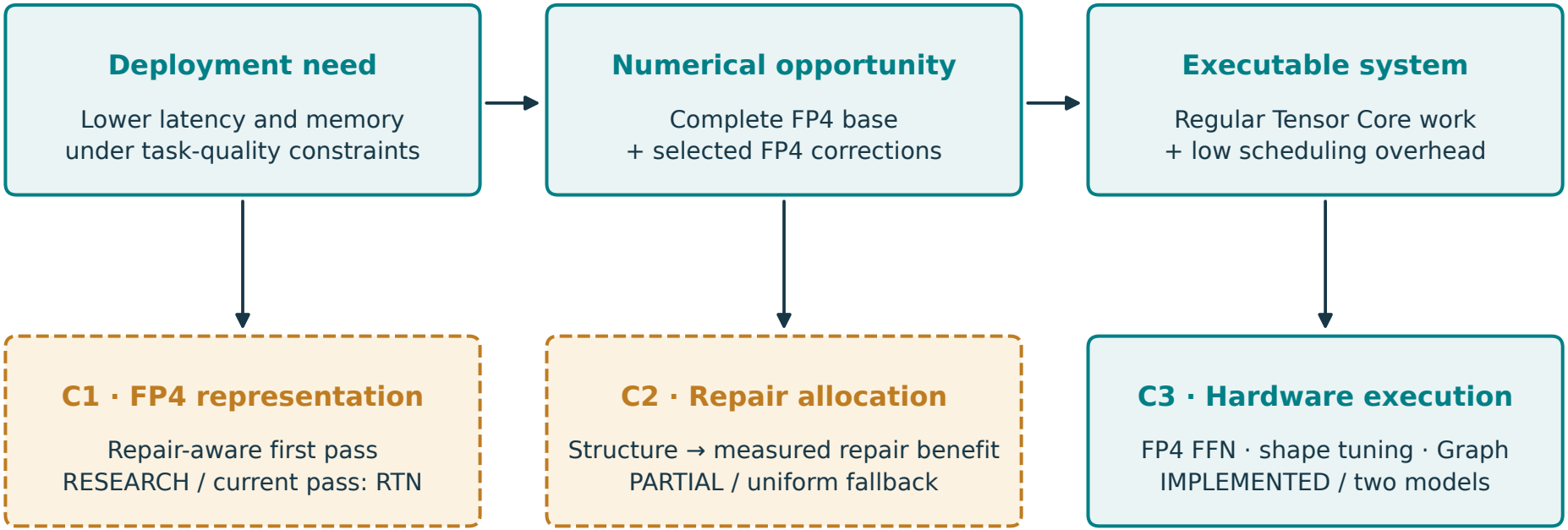

Validation target: complete-trajectory quality × latency × memory; energy remains unmeasured.